

MS 6015: Multivariate Statistical Methods for Business

Vectors and Matrices¹

¹Reference: Johnson and Wichern: Chapter 2

Outline

Vectors

- Notion of Distance

- Vector Projection

- Linear Independent Vectors and Basis

Matrices

- Introduction and Basic Notation

- Eigen Values and Eigen Vectors

- Rotation

- Quadratic Form and Positive Definite Matrices

- Distance and Quadratic forms

Random Vectors and Matrices

Learning Outcomes

- ▶ Interpret multivariate data using **vector geometry**
- ▶ Explain dot products, norms, angles, and distances geometrically
- ▶ Interpret a matrix as a **linear transformation**
- ▶ Understand **subspaces, span, rank**, and **projections**
- ▶ Connect covariance, eigenvectors and SVD to **PCA geometry**

Why Geometry for Multivariate Data?

Let $X \in \mathbb{R}^{n \times p}$ denote multivariate data:

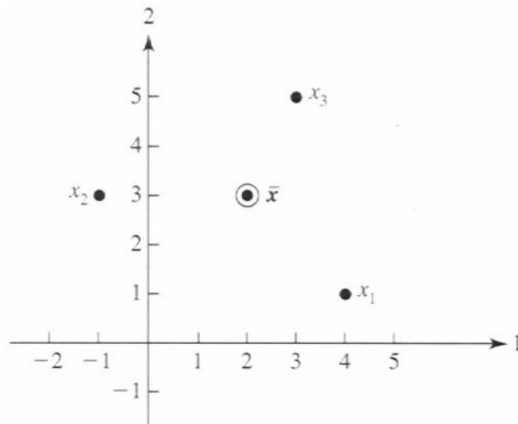
- ▶ **Row geometry:** each observation is a point $x_i \in \mathbb{R}^p$
- ▶ **Column geometry:** each variable is a vector in $\mathbb{R}^n$

Geometry powers:

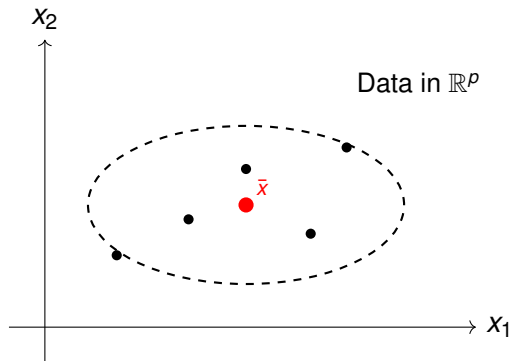
- ▶ similarity and distance
- ▶ clustering and embeddings
- ▶ regression as projection
- ▶ PCA as best low-dimensional representation

Representation of vector in 2- dimensional space

Example: $x_1 = \begin{pmatrix} 4 \\ 1 \end{pmatrix}, x_2 = \begin{pmatrix} -1 \\ 3 \end{pmatrix}, x_3 = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$



Multivariate Data as a Cloud of Points



A Single Variable as a Vector (and its Length)

A single variable measured across n observations is a column vector

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n, \quad \|x\| = \sqrt{x^\top x}.$$

- ▶ A vector has both length and direction.
- ▶ The length of the vector x is defined to be $L_x = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$.

Variance as Vector Length (Geometry in $\mathbb{R}^n$)

A single variable across n observations is a vector $x \in \mathbb{R}^n$. Centering removes the mean component:

$$x_c = x - \bar{x}\mathbf{1}.$$

The sample variance is the squared length of the centered vector (scaled):

$$s^2 = \frac{1}{n-1} \|x_c\|^2 \quad \Longleftrightarrow \quad \|x_c\| = \sqrt{(n-1)s^2}.$$

Scalar Multiplication

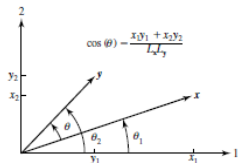
- ▶ What happens when we multiply a vector x by a scalar λ ?

$$\lambda \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix} = \begin{pmatrix} \lambda x_1 \\ \lambda x_2 \\ \vdots \\ \lambda x_m \end{pmatrix}$$

- ▶ effect is either to stretch or shrink
- ▶ concept of normalization.

Concept of angle

- ▶ Consider two vectors, $\mathbf{x}$, $\mathbf{y}$ in a plane and the angle θ between them.
- ▶ θ can be represented as difference of angles θ_1 and θ_2



$$\cos(\theta) = \frac{x_1 y_1 + x_2 y_2}{L_x L_y}$$

Dot Product and Angle

For vectors $x, y \in \mathbb{R}^n$:

$$x^\top y = \|x\| \|y\| \cos \theta$$

- ▶ $x^\top y$ measures alignment
- ▶ $\cos \theta$ gives **cosine similarity**
- ▶ if $\theta = 90^\circ$, vectors are orthogonal

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Correlation link: after centering and scaling variables, correlation $\approx$ cosine similarity.

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Norms and Distances

Norm: $\|x\|_2 = \sqrt{x^\top x}$

Euclidean distance:

$$d(x, y) = \|x - y\|$$

- ▶ distance defines clustering and nearest-neighbor structure
- ▶ scaling of variables changes distances dramatically

Norms and Distances

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Preview: Mahalanobis distance rescales space using covariance.

Distance

- ▶ Most multivariate techniques are based on the simple concept of distance.
- ▶ If we consider the point $P = (x_1, x_2)$ in a plane, the straight line distance, $d(O, P)$, from P to the origin $O = (0, 0)$ is

$$d(O, P) = \sqrt{x_1^2 + x_2^2}$$

- ▶ If we consider the point $P = (x_1, x_2, \dots, x_n)$ in a plane, the straight line distance, $d(O, P)$, from P to the origin $O = (0, 0, \dots, 0)$ is

$$d(O, P) = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

Distance Contd.

- ▶ If we consider the point $P = (x_1, x_2, \dots, x_n)$ and $Q = (y_1, y_2, \dots, y_n)$ in a plane, the straight line distance, $d(P, Q)$, from P to the point Q is

$$d(PQ) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2}$$

- ▶ Straight line or Euclidean distance is unsatisfactory for most statistical purposes. This is because each coordinate contributes equally to the calculation of Euclidean distance.

Statistical Distance

- ▶ Divide each coordinate by the sample standard deviation.
- ▶ We have standardized coordinates $x_1^* = x_1 / \sqrt{s_{11}}$ and $x_2^* = x_2 / \sqrt{s_{22}}$
- ▶ The standardised coordinates are on equal footing with one another.
- ▶ The Statistical distance of the point $P(x_1, x_2)$ from the origin $O = (0, 0)$ can be computed as

$$d(O, P) = \sqrt{\frac{x_1^2}{s_{11}} + \frac{x_2^2}{s_{22}}}$$

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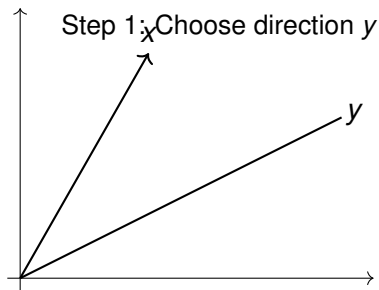
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Vector Projection in Two Dimensions

Definition: The projection of vector $\mathbf{x}$ onto vector $\mathbf{y}$ is given by:

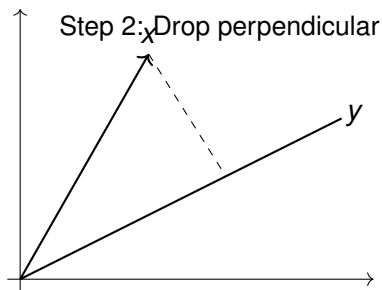
$$\text{proj}_{\mathbf{y}} \mathbf{x} = \frac{\mathbf{x}^T \cdot \mathbf{y}}{\mathbf{y}^T \cdot \mathbf{y}} \mathbf{y} = \frac{\mathbf{x}^T \cdot \mathbf{y}}{L_y} \frac{\mathbf{y}}{L_y}$$



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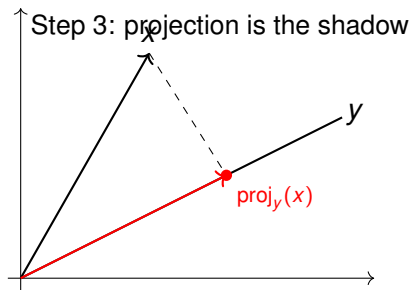
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Notion of linear dependence

- ▶ A pair of vectors $\mathbf{x}$ and $\mathbf{y}$ of the same dimension is said to be linearly dependent if there exist constants λ_1 and λ_2 , both not zero such that

$$\lambda_1 \mathbf{x} + \lambda_2 \mathbf{y} = \mathbf{0}$$

- ▶ A set of vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ of the same dimension is said to be linearly dependent if there exist constants $\lambda_1, \lambda_2, \dots, \lambda_k$ not all zero such that

$$\lambda_1 \mathbf{x}_1 + \lambda_2 \mathbf{x}_2 + \dots + \lambda_k \mathbf{x}_k = \mathbf{0}$$

- ▶ Vectors of same dimension that are not linearly dependent are said to be linearly independent.

Notion of Basis

- ▶ Any set of m linearly independent vectors is called a basis for the vector space of all m -tuples of real numbers.
- ▶ Every vector in $\mathbb{R}^m$ can be represented as a linear combination of a finite basis.

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Matrix

- ▶ A matrix is any rectangular array of real numbers.

- ▶ $\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1p} \\ a_{21} & a_{22} & \dots & a_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{np} \end{pmatrix}$

Matrix Operations

- ▶ Transpose of a matrix
- ▶ Sum of two matrices
- ▶ Matrix Multiplication

Special Matrices

- ▶ Symmetric Matrices: Square matrix is said to be symmetric if $A = A'$ or $a_{ij} = a_{ji}$ for all i and j .
- ▶ Identity Matrix: I
- ▶ Orthogonal Matrix: $QQ' = Q'Q = I$ or $Q' = Q^{-1}$

Covariance Matrix: Geometry of Variables

With centered data matrix X :

$$S = \frac{1}{n-1} X^{\top} X$$

- ▶ diagonal = variances (spread along axes)
- ▶ off-diagonal = covariances (co-movement)

Covariance Matrix: Geometry of Variables

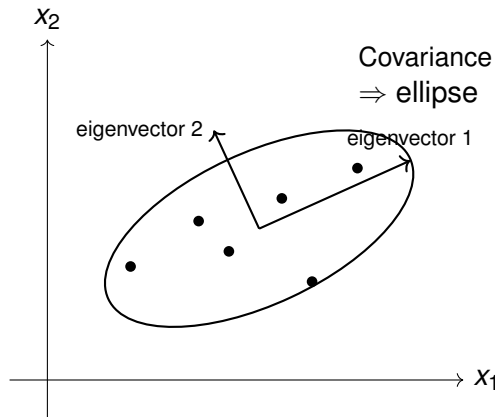
With centered data matrix X :

$$S = \frac{1}{n-1} X^{\top} X$$

- ▶ diagonal = variances (spread along axes)
- ▶ off-diagonal = covariances (co-movement)

Covariance induces ellipsoidal geometry in feature space.

Covariance Geometry: Data Ellipse



Eigenvectors of S align with ellipse axes; eigenvalues set axis lengths.

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Eigen Values and Eigen Vectors

- ▶ A square matrix A has an eigenvalue λ with corresponding eigenvector $z \neq 0$ if

$$Az = \lambda z$$

- ▶ The eigenvalues of A are the solution to $|A - \lambda I| = 0$.
- ▶ A normalized eigenvector (of unit length) is denoted by e .
- ▶ A $k \times k$ matrix A has k pairs of eigenvalues and eigenvectors

$$\lambda_1, e_1; \lambda_2, e_2; \dots, \lambda_k, e_k$$

where $e_i' e_i = 1$ and $e_i' e_j = 0$ and the eigenvectors are unique up to a change in sign unless two or more eigenvalues are equal.

Example

▶ $\begin{bmatrix} 1 & -5 \\ -5 & 1 \end{bmatrix}$



$$(1 - \lambda)^2 - 25 = 0 \iff \lambda^2 - 2\lambda - 24 = 0.$$

- ▶ Eigen values are $\lambda_1 = 6$ and $\lambda_2 = -4$ with corresponding eigen vectors being

$$e_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 1 \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \text{ and } e_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 1 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

Spectral Decomposition

- ▶ Eigenvalues and eigenvectors will play an important role in this course.
 - ▶ For example, principal components are based on the eigenvalues and eigenvectors of sample covariance matrices.
- ▶ The spectral decomposition of a $k \times k$ symmetric matrix A is

$$A = \lambda_1 \mathbf{e}_1 \mathbf{e}_1' + \lambda_2 \mathbf{e}_2 \mathbf{e}_2' + \dots + \lambda_k \mathbf{e}_k \mathbf{e}_k'$$
$$= [\mathbf{e}_1 \quad \mathbf{e}_2 \quad \dots \quad \mathbf{e}_k] \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_k \end{bmatrix} \begin{bmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \\ \vdots \\ \mathbf{e}_k \end{bmatrix} = P \Lambda P'$$

where $\lambda_1, \lambda_2, \dots, \lambda_k$ are the eigen values of A and $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_k$ are the associated normalized eigen vectors.

Example 1

Let

$$A = \begin{bmatrix} 1 & -5 \\ -5 & 1 \end{bmatrix}.$$

Its eigenpairs are

$$\lambda_1 = 6, \quad \mathbf{e}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \lambda_2 = -4, \quad \mathbf{e}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Define the orthogonal matrix $P = [\mathbf{e}_1 \ \mathbf{e}_2]$ and diagonal matrix

$$P = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \quad \Lambda = \begin{bmatrix} 6 & 0 \\ 0 & -4 \end{bmatrix}.$$

Then the spectral decomposition is

$$A = P \Lambda P^T.$$

Equivalently,

$$A = 6 \mathbf{e}_1 \mathbf{e}_1^T - 4 \mathbf{e}_2 \mathbf{e}_2^T = 6 \left(\frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \right) - 4 \left(\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \right).$$

Example 2

Consider the symmetric matrix

$$A = \begin{bmatrix} 13 & -4 & 2 \\ -4 & 13 & -2 \\ 2 & -2 & 10 \end{bmatrix}$$

- ▶ The eigen values are $\lambda_1 = 9$, $\lambda_2 = 9$ and $\lambda_3 = 18$
- ▶ The corresponding eigen vectors

$$e_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} \frac{1}{\sqrt{18}} \\ -\frac{1}{\sqrt{18}} \\ -\frac{4}{\sqrt{18}} \end{bmatrix}, e_3 = \begin{bmatrix} \frac{2}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{bmatrix}$$

$$\begin{pmatrix} 13 & -4 & 2 \\ -4 & 13 & -2 \\ 2 & -2 & 10 \end{pmatrix} = \begin{pmatrix} \frac{2}{3} & \frac{\sqrt{2}}{2} & \frac{-\sqrt{2}}{6} \\ \frac{-2}{3} & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{6} \\ \frac{1}{3} & 0 & \frac{2*\sqrt{2}}{3} \end{pmatrix} \begin{pmatrix} 18 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix} \begin{pmatrix} \frac{2}{3} & \frac{\sqrt{2}}{2} & \frac{-\sqrt{2}}{6} \\ \frac{-2}{3} & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{6} \\ \frac{1}{3} & 0 & \frac{2*\sqrt{2}}{3} \end{pmatrix}$$

Why Spectral Decomposition

- ▶ Spectral Decomposition is an important analytical tool.
- ▶ Very useful to demonstrate statistical results.
- ▶ Useful in matrix explanation of distance

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Matrix as a Geometric Operator

A matrix $A \in \mathbb{R}^{p \times p}$ defines a transformation:

$$x \mapsto Ax$$

Geometric effects:

- ▶ scaling (stretching)
- ▶ rotation

Key idea:

Matrices transform *space*.

Orthonormal Matrices (Rotations)

A matrix Q is orthonormal if:

$$Q^T Q = I$$

Properties:

- ▶ preserves length: $\|Qx\| = \|x\|$
- ▶ preserves angles: $\angle(Qx, Qy) = \angle(x, y)$
- ▶ represents rotation

Orthonormal Matrices (Rotations)

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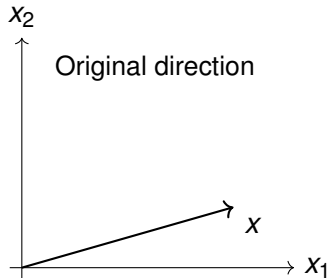
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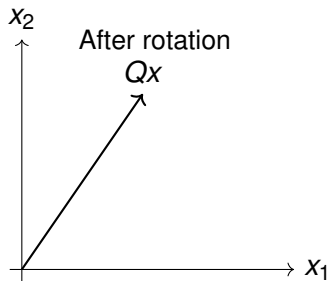
PCA uses rotations: it rotates data to new coordinate axes.

Rotation as a Matrix



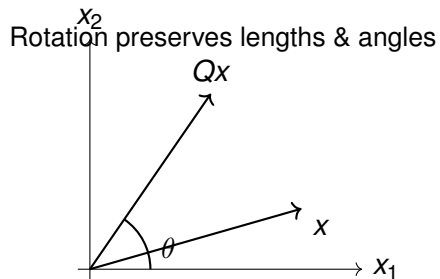
$$Q = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Rotation as a Matrix



Qx is a rotated version of x

Rotation as a Matrix



$$\|Qx\| = \|x\|, \quad (Qx)^{\top}(Qy) = x^{\top}y$$

Why Rotations Preserve Length and Angles

A 2D rotation by angle θ is represented by the orthonormal matrix

$$Q = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}, \quad Q^\top Q = I.$$

Length preservation: for any vector $x \in \mathbb{R}^2$,

$$\|Qx\|^2 = (Qx)^\top (Qx) = x^\top Q^\top Qx = x^\top x = \|x\|^2.$$

Angle preservation: for any $x, y \in \mathbb{R}^2$,

$$(Qx)^\top (Qy) = x^\top Q^\top Qy = x^\top y \quad \Rightarrow \quad \cos \angle(Qx, Qy) = \cos \angle(x, y).$$

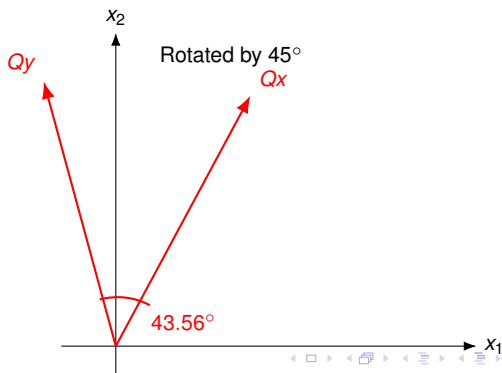
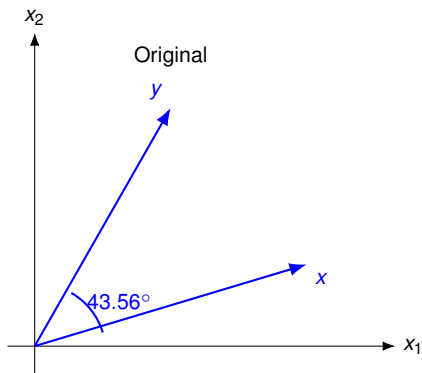
illustration

Let

$$x = \begin{bmatrix} 4 \\ 1.2 \end{bmatrix}, \quad y = \begin{bmatrix} 2 \\ 3.5 \end{bmatrix}, \quad Q = \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{bmatrix}.$$

Then

$$\angle(x, y) = \angle(Qx, Qy) \approx 43.56^\circ.$$



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Quadratic Form

- ▶ For a $k \times k$ symmetric matrix A and a vector $x = (x_1, x_2, \dots, x_k)'$ the quantity $x'Ax$ is called a *quadratic form*
- ▶ Note that
$$x'Ax = \sum_{i=1}^k \sum_{j=1}^k a_{ij}x_i x_j$$
- ▶ If $x'Ax \geq 0$ for any vector x , both A and the quadratic form are said to be non-negative definite.
- ▶ If $x'Ax > 0$ for any vector $x \neq 0$, both A and the quadratic form are said to be positive definite.

Example

- ▶ Consider $3x_1^2 + 2x_2^2 - 2\sqrt{2}x_1x_2$.
 - ▶ The quadratic form on matrix notation is

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 3 & -\sqrt{2} \\ -\sqrt{2} & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x'Ax$$

- ▶ Verify Eigen values are $\lambda_1 = 4$ and $\lambda_2 = 1$.
 - ▶ Spectral Decomposition: $A = 4e_1e_1' + e_2e_2'$
- ▶ Pre multiplication and postmultiplication of A by x' and x gives
 - ▶ $x'Ax = 4x'e_1e_1'x + x'e_2e_2'x = 4y_1^2 + y_2^2 \geq 0$

Positive Definiteness of A

- ▶ We can show y_1 and y_2 are not both zero.
 - ▶ $\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} e_1' \\ e_2' \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}; y = Ex$
 - ▶ E is orthogonal and has an inverse; hence $x = E^{-1}y. 0 \neq x = E^{-1}y \implies y \neq 0$
- ▶ Using spectral decomposition, we can easily show that a $k \times k$ symmetric matrix A is positive definite matrix if and only if every eigen value of A is positive.
- ▶ A is a nonnegative definite matrix if and only if all of its eigen values are greater than or equal to zero.

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Distance

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$$d(O, P) = \sqrt{x_1^2 + x_2^2}$$

- ▶ If we consider the point $P = (x_1, x_2, \dots, x_p)$ in a plane, the straight line distance, $d(O, P)$, from P to the origin $O = (0, 0, \dots, 0)$ is

$$d(O, P) = \sqrt{x_1^2 + x_2^2 + \dots + x_p^2}$$

Distance Contd.

- ▶ If we consider the point $P = (x_1, x_2, \dots, x_p)$ and $Q = (y_1, y_2, \dots, y_p)$ in a plane, the straight line distance, $d(P, Q)$, from P to the origin Q is

$$d(PQ) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_p - y_p)^2}$$

- ▶ Straight line or Euclidean distance is unsatisfactory for most statistical purposes. This is because each coordinate contributes equally to the calculation of Euclidean distance.

Statistical Distance

- ▶ Divide each coordinate by the sample standard deviation.
- ▶ We have standardized coordinates $x_1^* = x_1 / \sqrt{s_{11}}$ and $x_2^* = x_2 / \sqrt{s_{22}}$
- ▶ The standardised coordinates are on equal footing with one another.
- ▶ The Statistical distance of the point $P(x_1, x_2)$ from the origin $O = (0, 0)$ can be computed as

$$d(O, P) = \sqrt{\frac{x_1^2}{s_{11}} + \frac{x_2^2}{s_{22}}}$$

In general, the statistical distance of the point $P = (x_1, x_2)$ from the fixed point $Q = (y_1, y_2)$ for situations in which the variables are correlated has the general form

$$d(P, Q) = \sqrt{a_{11}(x_1 - y_1)^2 + 2a_{12}(x_1 - y_1)(x_2 - y_2) + a_{22}(x_2 - y_2)^2}$$

where $a_{11} = \frac{1}{s_{11}}$, $a_{22} = \frac{1}{s_{22}}$ and a_{12} is covariance inverse.

$$d(P, Q) = \sqrt{a_{11}(x_1 - y_1)^2 + 2a_{12}(x_1 - y_1)(x_2 - y_2) + a_{22}(x_2 - y_2)^2}$$

can be expressed as $\sqrt{(x - y)' A (x - y)}$ where $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{bmatrix}$

Distance and Quadratic Forms

- ▶ For $x = (x_1, x_2, \dots, x_p)'$ and a $p \times p$ positive definite matrix A ,

$$distance^2 = x'Ax > 0$$

where $x \neq 0$. Thus, a positive definite quadratic form can be interpreted as a squared distance of x from the origin and vice versa.

- ▶ the squared distance from x to a fixed point μ is given by the quadratic form

$$(x - \mu)'A(x - \mu)$$

Distance and Quadratic Forms

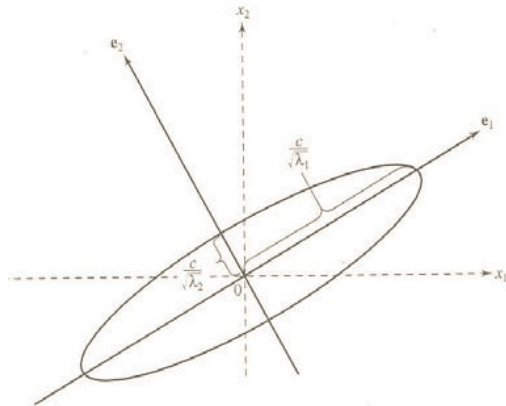
- ▶ We can interpret distance in terms of eigenvalues and eigenvectors of A as well. Any point x at constant distance c from the origin satisfies

$$x'Ax = x' \left(\sum_{j=1}^p \lambda_j e_j e_j' \right) x = \sum_{j=1}^p \lambda_j (x' e_j)^2 = c^2$$

the expression for an ellipsoid in p dimensions.

- ▶ All points at distance c lie on an ellipsoid with axes in the directions of the eigenvectors and with lengths proportional to $\lambda_j^{-1/2}$

Figure



Square Root Matrix

- ▶ the Spectral Decomposition allows us to express the inverse of a square matrix in terms of its eigen values and eigen vectors, and this leads to a useful square-root matrix.
- ▶ The square-root matrix, of a positive definite matrix A ,

$$A^{1/2} = \sum_{i=1}^k \sqrt{\lambda_i} \mathbf{e} \mathbf{e}' = P \Lambda^{1/2} P'$$

Random Vectors and Matrices

- ▶ A random vector is a vector whose elements are random variables. Similarly, a random matrix is a matrix whose elements are random variables.
- ▶ Let $\mathbf{X} = \{\mathbf{X}_{ij}\}$ be an $n \times p$ random matrix. Then the expected value of $\mathbf{X}$, denoted by $E(\mathbf{X})$ is the matrix of numbers (if they exist)

$$\begin{bmatrix} E(X_{11}) & E(X_{12}) & \dots & E(X_{1p}) \\ E(X_{21}) & E(X_{22}) & \dots & E(X_{2p}) \\ \vdots & \vdots & \ddots & \vdots \\ E(X_{n1}) & E(X_{n2}) & \dots & E(X_{np}) \end{bmatrix}$$

Exercise

- ▶ Suppose $p = 2$ and $n = 1$, and consider the random vector $\mathbf{X}' = [\mathbf{X}_1, \mathbf{X}_2]$
- ▶ Let the discrete random variable X_1 have the following probability function:

X_1	-1	0	1
$P(X_1 = x_1)$	0.3	0.3	0.4

- ▶ Let the discrete random variable X_2 have the following probability function:

X_2	0	1
$P(X_2 = x_2)$	0.8	0.2

Computing the Covariance Matrix

Find the covariance matrix for the two random variables when their joint probability function is represented by the entries

X_1 / X_2	0	1	$P(X_1 = x_1)$
-1	0.24	0.06	0.3
0	0.16	0.14	0.3
1	0.40	0	0.4
$P(X_2 = x_2)$	0.8	0.2	

- ▶ Mean vector: $E(\mathbf{X}) = \mu = \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix}$
- ▶ Variance- covariance matrix:

$$\Sigma = E(\mathbf{X} - \mu)(\mathbf{X} - \mu)' = \begin{bmatrix} 0.69 & -0.08 \\ -0.08 & 0.16 \end{bmatrix}$$

Summary Statistics

Summary numbers to assess the information contained in data Basic descriptive statistics

- ▶ Sample mean
- ▶ Sample variance
- ▶ Sample standard deviation
- ▶ Sample covariance
- ▶ Sample correlation coefficient